

HEALTHCONNECT CLINIC

WEEK 8 — FINAL ANALYTICS & DECISION SUPPORT PACKAGE

Experience Lab Internship Programme

Data Analytics Track

Intern: Lisemanyane Mporoane

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and
AI

Organisation: AnalystLab Africa

1. Executive Summary

The HealthConnect Clinic Data Analytics work focused on understanding appointment attendance, no-show behaviour and factors associated with missed appointments.

Weeks 5–6 established the exploratory and advanced analytical foundation, while Week 7 focused on testing, KPI validation, dashboard filter testing and refinement. Week 8 transforms those validated outputs into a final decision-support package for the wider HealthConnect solution.

The final analysis was based on 5,000 appointment records. The validated overall no-show rate is 48.46%, representing 2,423 no-show appointments.

The strongest analytical patterns identified were associated with booking lead time and previous no-show history. No-show rates increased from 29% for appointments booked 0–9 days in advance to 68% for appointments booked 50–59 days in advance. Previous no-show history also showed an increasing pattern, from 44% among patients with no previous no-shows to 68% among patients with three previous no-shows.

Reminder-related variables also showed differences in observed no-show rates. Appointments without a reminder had a 51% no-show rate, compared with 47% where a reminder was sent. By reminder channel, the observed rates were 51% for no channel, 50% for WhatsApp, 48% for Email and 46% for SMS.

The analysis does not establish that any factor causes a patient to miss an appointment. The findings identify patterns that HealthConnect can investigate further and potentially use to support targeted appointment-management strategies.

The final analytics component is ready for integration with the wider HealthConnect solution.

2. Project Background

HealthConnect Clinic is seeking to improve appointment attendance and healthcare administrative support through data, analytics, predictive methods and AI. The Data Analytics

track focused on understanding appointment outcomes and identifying patterns that can support operational decisions.

- How many appointments are being missed?
- What is the overall no-show rate?
- Which appointment characteristics are associated with higher no-show rates?
- Are booking lead times associated with attendance?
- Does previous no-show history reveal an important pattern?
- What patterns exist around appointment reminders?
- Are demographic differences substantial?
- Which findings are reliable enough to support business decisions?
- How can the findings support the wider HealthConnect solution?

3. Week 7 → Week 8 Transition

Item	Response
Main Week 7 output	A tested and refined HealthConnect analytics dashboard containing validated KPIs, analytical findings, dashboard filters and testing evidence.
Most important testing result	The overall no-show KPI was validated at 48.46% (2,423 no-shows out of 5,000 appointments).
Major issue discovered	Some very high no-show percentages were based on extremely small groups, particularly patients with four or five previous no-shows.
What was refined	Rates were interpreted alongside counts; the 60+ booking-lead-days plateau was clarified; KPI calculations and dashboard filters were tested.
Component now ready	HealthConnect Final Analytics & Decision Support component.
Still needs integration	Validated analytics findings need to connect

	with predictive, technical, GenAI and project-management components.
Tracks involved	Data Analytics, Data Science, ML Engineering, Generative AI and Project Management.
Final presentation focus	Demonstrate how validated analytics identify no-show patterns, translate into business recommendations and support the wider HealthConnect solution.

4. Final Integration Readiness

Requirement	Final Response
What is my final component?	HealthConnect Final Analytics & Decision Support Package.
What problem does it address?	It helps HealthConnect understand appointment attendance and no-show patterns so management can identify areas requiring attention.
What is its current status?	Ready for final integration.
What did I improve during Week 7?	KPI calculations, analytical findings, dashboard filters, sample-size interpretation, visual interpretation and testing evidence.
Which other tracks does it depend on?	Data Science, ML Engineering and GenAI may provide outputs or requirements relevant to final integration.
Which tracks depend on my output?	Data Science and Project Management can use validated analytical findings; other tracks can use relevant analytics when connecting their components.
What output will I provide?	Final dashboard/report, KPIs, validated findings, business insights, recommendations, limitations and

	presentation material.
What output do I need from another track?	Relevant final predictive findings and technical requirements where applicable.
What evidence demonstrates readiness?	Week 7 testing evidence, validated KPI calculations, dashboard outputs, filter testing and documented refinements.
What limitations remain?	Missing values, small sample sizes, observational data, historical-data limitations and an independent KPI discrepancy requiring reconciliation.

5. Dataset and Analytical Approach

The HealthConnect dataset contains 5,000 appointment records. Variables included appointment, patient, reminder and outcome information such as Age, Gender, Age Group, Appointment Type, Booking Lead Days, Previous No-Shows, Reminder Sent, Reminder Channel, Distance to Clinic, Waiting Time and Appointment Outcome.

The analysis used Power BI for calculated measures, visualisations, segmentation and dashboard filters.

Data preparation → KPI calculation → Exploratory analysis → Advanced segmentation → Testing → Validation → Dashboard refinement → Business insights → Recommendations

6. Data Quality Review

6.1 Appointment IDs

Appointment IDs were checked for duplicates. No duplicate Appointment IDs were identified.

6.2 Patient IDs

Repeated Patient IDs were not treated as duplicate records because a patient can have multiple appointments.

6.3 Categorical Variables

Gender, Age Group, Appointment Type, Reminder Sent, Reminder Channel and Appointment Outcome were reviewed for consistency.

6.4 Missing Distance Values

There were 90 missing Distance to Clinic values. These were not replaced with zero.

6.5 Missing Waiting-Time Values

There were 60 missing Waiting Time values. These were retained as missing rather than converted to zero.

6.6 Overall Data Quality Position

The available data was sufficient for the main attendance/no-show analysis, but missing values and the historical nature of the dataset remain limitations.

7. Final KPI Validation

KPI	Final Value
Total Appointments	5,000
No-Show Appointments	2,423
No-Show Rate	48.46%
Cancellation Rate	5.26%
Attendance Rate	46.28%
Non-Attendance Rate	53.72%

No-Show Rate validation: $2,423 \div 5,000 \times 100 = 48.46\%$. The main outcome rates reconcile to 100%: 46.28% Attendance + 48.46% No-Show + 5.26% Cancellation = 100%.

The dashboard may display approximately 0.48 depending on formatting, but the validated percentage is 48.46%.

8. Final Analytical Findings

8.1 Booking Lead Days

Booking Lead Days	Appointments	No-Show Rate
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0–9	817	29%
10–19	865	37%
20–29	810	44%
30–39	843	51%
40–49	826	62%
50–59	745	68%
60+	94	68%

The observed no-show rate increased substantially through the 50–59-day group and remained at 68% for 60+, so the analysis should not claim continued increase beyond 60 days.

8.2 Previous No-Show History

Previous No-Shows	Appointments	No-Show Rate
0	2,921	44%
1	1,548	53%
2	438	59%
3	78	68%
4	12	67%
5	3	100%

The results show an increasing pattern from zero to three previous no-shows. The 4- and 5-previous-no-show groups are very small and should be interpreted cautiously.

8.3 Reminder Status

Reminder Sent	Appointments	No-Show Rate
No	1,366	51%
Yes	3,644	47%

Appointments where a reminder was recorded had a lower observed no-show rate. This observational relationship does not establish causation.

8.4 Reminder Channel

Reminder Channel	Appointments	No-Show Rate
None	1,366	51%
WhatsApp	1,101	50%

Email	533	48%
SMS	2,000	46%

SMS had the lowest observed no-show rate among the listed reminder channels. This is an area for further investigation rather than proof that SMS causes lower no-show rates.

8.5 Age Group

Age Group	Appointments	No-Show Rate
18–24	564	50%
25–34	783	51%
35–44	819	48%
45–54	793	48%
55–64	800	51%
65+	1,241	45%

The 65+ group had the largest appointment volume but the lowest observed no-show rate among the age groups. This demonstrates why raw appointment counts should not be used alone to identify higher-risk groups.

8.6 Gender

Gender	Appointments	No-Show Rate
Female	2,488	48%
Male	2,404	49%
Prefer not to say	108	44%

The difference between female and male no-show rates was approximately one percentage point. The smaller 'Prefer not to say' group should be interpreted cautiously.

9. Combined Analysis

A combined Previous No-Shows × Booking Lead Days analysis was used to determine whether the individual patterns remained visible when analysed together.

Previous No-Shows	Booking Lead Days	Appointments	No-Show Rate
0	0–9	483	24%
0	50–59	423	62%
0	60+	51	65%
3	60+	2	100%
4	40–49	2	100%
4	50–59	2	100%
5	All relevant	3	100%

	appointments		
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The booking-lead-time pattern remained visible among patients with zero previous no-shows. Extreme 100% cells were based on only two or three appointments and should not be treated as stable estimates.

10. Final Business Insights

- Booking lead time is an important observed risk signal: the no-show rate increased from 29% at 0–9 days to 68% at 50–59 days.
- Previous no-show history is an important analytical variable: the rate increased from 44% among patients with no previous no-shows to 68% among those with three previous no-shows.
- Reminder status shows an observed difference: 51% without a recorded reminder versus 47% where a reminder was recorded.
- Reminder channels show different observed rates: SMS 46%, Email 48%, WhatsApp 50% and None 51%.
- Demographic volume does not automatically indicate risk: the 65+ group had the largest volume but a 45% no-show rate.
- Small groups require caution: some 100% rates were based on only two or three appointments.

11. Updated Business Recommendations

1. Strengthen confirmation for long-lead appointments, particularly those booked 50 or more days in advance.
2. Use previous no-show history to support additional, non-punitive follow-up.
3. Continue improving the reminder process and monitor outcomes.
4. Further evaluate reminder channels, including SMS, using more controlled analysis where possible.
5. Develop a future risk-based strategy using Previous No-Shows, Booking Lead Days, Reminder Sent, Reminder Channel, Age Group, Gender and Appointment Type.
6. Monitor rates and counts together to avoid over-interpreting small groups.
7. Maintain regular KPI validation against the underlying appointment data.

12. Final Dashboard and Visualisation Design

- Overall No-Show KPI
- No-Show Rate by Booking Lead Days
- No-Show Rate by Previous No-Shows
- No-Show Rate by Reminder Sent
- No-Show Rate by Reminder Channel
- No-Show Rate by Age Group

- No-Show Rate by Gender
- Previous No-Shows × Booking Lead Days matrix
- Relevant filters and slicers

Week 7 refinement focused on testing filters, checking returned counts and rates, adding counts alongside percentages, improving interpretation of small groups and avoiding unsupported conclusions from raw counts.

13. Dashboard Validation Evidence

Test	Validated Result
Age Group = 65+	1,241 appointments; gender counts 621 + 592 + 28 = 1,241
Booking Lead Days = 50–59	745 appointments; 68% no-show rate
Reminder Sent = No	1,366 appointments; 51% no-show rate
Reminder Channel = SMS	2,000 appointments; 46% no-show rate
Clear filters	Returned to 5,000 appointments and approximately 0.48 displayed no-show rate

These tests demonstrated that the dashboard responded appropriately to selected filters. The exact validated overall KPI remains 48.46%.

14. Cross-Track Contribution

14.1 Data Analytics → Data Science

Validated findings can support future predictive modelling. Candidate variables include Previous No-Shows, Booking Lead Days, Reminder Sent, Reminder Channel, Age Group, Gender and Appointment Type.

14.2 Data Analytics → Project Management

The final analytics component provides validated KPIs, key findings, business insights, recommendations and limitations for project decisions and the final presentation.

14.3 Integration with ML Engineering

Analytics findings can provide business and data context for technical workflow development. The analytics component does not independently implement the ML workflow.

14.4 Integration with Generative AI

Validated analytics findings can provide relevant business context for the GenAI component where appropriate, subject to its defined capabilities and limitations.

15. Final HC-POD Integration

Business Problem → HealthConnect Data → Data Analytics → Validated Findings & Business Insights → Data Science / Predictive Analysis → ML Engineering / Technical Workflow → Generative AI / Administrative Support → Project Management & Decision-Making → Final HealthConnect Solution

The Data Analytics component acts as an evidence layer connecting appointment data to business decisions and supporting interpretation of other technical components.

16. Final End-to-End HealthConnect Solution

Component	Contribution
The Problem	HealthConnect needs to better understand and manage appointment non-attendance.
The Data	5,000 appointment records containing appointment, patient, reminder and outcome variables.
The Analytics	Identifies patterns in booking lead time, previous no-shows, reminders and demographic segments.
Predictive Component	Data Science can investigate predictive relationships using validated analytical findings.
ML Engineering	Organises predictive outputs into a usable technical workflow where applicable.
GenAI	Supports the administrative experience within its defined capabilities and limitations.
Project Management	Coordinates workstreams, dependencies, deliverables and final presentation.
Final Solution	Connects data, analytics, predictive analysis, technical workflow, AI support and business decisions.

17. Limitations

- 90 missing Distance values.
- 60 missing Waiting Time values.
- Small segments can produce unstable extreme percentages.
- The analysis is observational and does not establish causation.
- Historical data may not perfectly represent future patient behaviour.
- Reminder groups may differ in other characteristics, creating potential confounding.

- An independent analysis reported a different overall no-show figure; the validated calculation in this report is 48.46%, and the discrepancy should be reconciled before one HC-POD-wide KPI is published.
- The dashboard's 0.48 display should be formatted as a percentage for clearer communication.

18. Remaining Analytical Issues

Issue	Status
KPI calculation validation	Validated
Previous No-Shows pattern	Validated with small-group caution
Booking Lead Days pattern	Validated with 60+ plateau noted
Reminder Sent	Validated
Reminder Channel	Validated
Age Group	Validated
Gender	Validated
Dashboard filters	Tested and passed
Small sample segments	Documented limitation
Missing Distance values	Remaining limitation
Missing Waiting Time values	Remaining limitation
Causal interpretation	Limitation documented
Independent 51% vs 48.46% KPI discrepancy	Requires reconciliation
Dashboard percentage formatting	Minor refinement opportunity

19. Week 8 Final Analytics Status

COMPONENT STATUS: READY FOR FINAL INTEGRATION

The main analytical work has been developed, tested, validated, refined, documented and translated into business insights and recommendations. Remaining work is primarily to connect the analytics outputs with the wider HealthConnect multidisciplinary solution.

20. Executive Summary for Management

HealthConnect's dataset contains 5,000 appointment records. The validated overall no-show rate is 48.46%, representing 2,423 no-show appointments.

The strongest observed patterns were associated with booking lead time and previous no-show history. Appointments booked 0–9 days in advance had a 29% no-show rate, while appointments booked 50–59 days in advance had a 68% no-show rate.

Patients with no previous no-shows had a 44% no-show rate, increasing to 68% among patients with three previous no-shows.

Reminder status also showed an observed difference: 51% without a recorded reminder versus 47% where a reminder was recorded. Reminder-channel rates ranged from 46% for SMS to 51% for no recorded reminder channel.

The findings support further attention to long-lead appointments, previous no-show history, reminder optimisation and controlled testing of reminder channels. The analysis also demonstrates the importance of interpreting percentages together with appointment counts.

21. Final Presentation Contribution

8. Explain the HealthConnect appointment attendance problem.
9. Introduce the 5,000-record dataset and analytical approach.
10. Demonstrate the final dashboard.
11. Present the strongest findings: booking lead time, previous no-shows and reminder patterns.
12. Explain how Week 7 testing and refinement improved analytical reliability.
13. Present the business recommendations.
14. Explain limitations and the unresolved KPI discrepancy.
15. Explain how the analytics component connects to Data Science, ML Engineering, GenAI and Project Management.

22. Final Conclusion

The Week 8 Data Analytics work completes the transition from exploratory analysis to validated decision support. The analysis established a validated overall no-show rate of 48.46%. The strongest observed patterns concern booking lead time and previous no-show history, while reminder status and channel also show differences that warrant further investigation.

The Week 7 testing process demonstrated that analytical results should not be accepted simply because they appear visually convincing. KPI calculations, dashboard filters, segment sizes and unexpected results were checked before the findings were carried into the final solution.

The final dashboard and analytical package provide HealthConnect with a clearer evidence base for understanding appointment attendance and considering targeted interventions. The next stage is to integrate these validated analytical outputs with the other HC-POD components.

23. Appendix A — Final KPI Reference

Metric	Final Value
Total Appointments	5,000
No-Show Appointments	2,423
No-Show Rate	48.46%
Attendance Rate	46.28%
Cancellation Rate	5.26%
Non-Attendance Rate	53.72%
Missing Distance	90
Missing Waiting Time	60

24. Appendix B — Key Findings Reference

Analysis	Key Result
Booking Lead Days	29% → 68% across 0–9 to 50–59 days
Previous No-Shows	44% → 68% from 0 to 3 previous no-shows
Reminder Sent	51% without reminder vs 47% with reminder
Reminder Channel	SMS 46%, Email 48%, WhatsApp 50%, None 51%
Age	65+ had largest volume but 45% no-show rate
Gender	Female 48%, Male 49%, Prefer not to say 44%
Combined Analysis	Lead-time pattern remained visible among patients with zero previous no-shows

25. Appendix C — Evidence Checklist

- Final Power BI dashboard screenshot
- No-Show Rate by Previous No-Shows
- No-Show Rate by Booking Lead Days
- Reminder Sent analysis
- Reminder Channel analysis
- Age Group analysis
- Gender analysis
- Previous No-Shows × Booking Lead Days matrix
- KPI validation evidence
- Dashboard filter-testing evidence
- Clear-filter/reset evidence
- Before/after refinement evidence where applicable

- Week 7 testing documentation
- Cross-track integration evidence
- Final recommendations
- Limitations
- Individual presentation/video